
SOFTWARE REQUIREMENT SPECIFICATION

for
Loan Management System iOS
Application

Version History

Name	Date	Version
Sheetal R	03/04/2026	1.0

1. INTRODUCTION

1.1. APP VISION

The purpose of this document is to outline the requirements for a Loan Management System (LMS) mobile application built natively for iOS platforms.

1.2. PURPOSE OF THE APP

This project is not being developed for a specific client; however, it will be extensible enough to be adapted, customised, and deployed for integration into any financial institution or lending network. The primary objectives of the Loan Management System are to:

- Provide a comprehensive, user-friendly platform for managing the complete loan lifecycle — from application to closure
- Automate and streamline loan processing, approval workflows, EMI scheduling, and repayment tracking
- Enhance collaboration and communication between borrowers, loan officers, and administrators
- Ensure regulatory compliance, and auditability across all loan operations
- Deliver real-time insights, dashboards, and reports for informed decision-making

2. SCOPE OF THE PROJECT

2.1. HIGH LEVEL FEATURES

- Loan Application : Borrowers can submit loan applications with required details and documents directly from the app
 - Application Tracking : Real-time status tracking of loan applications through each stage of the workflow
 - EMI Calculator: Interactive EMI calculator supporting various loan types, tenures, and interest rate structures
 - Repayment Dashboard: Visual dashboard showing upcoming EMIs, payment history, outstanding balance, and overdue alerts
 - Document Upload : Secure upload and management of KYC documents, income proofs, and collateral papers
 - Loan Approval Workflow : Multi-level approval workflow for loan officers and managers with notification triggers
 - Notifications & Alerts: Push notifications for due dates, approval updates, disbursement confirmations, and overdue reminders
 - Reports & Analytics : Role-based reports covering portfolio health, repayment trends, and NPA analysis
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- Profile Management : Borrower and officer profile management with KYC status, credit score display, and history
- Messaging : In-app messaging between borrowers and loan officers for queries and document clarifications

2.2. TYPES OF USERS

Borrower — submits applications, tracks repayments, and communicates with officers

Loan Officer — reviews applications, manages customer queries, and initiates approval

Manager — oversees loan portfolio, approves or rejects officer recommendation

Admin — manages system configuration, user roles, and generates institutional reports

2.3. TECHNICAL SPECIFICATIONS

Devices: iPhone, iPad, OS: iOS 26+.

Frameworks: SwiftUI, Core ML, Vision, App Intents.

Architecture: MVVM with Swift Concurrency.

Security: Passkeys, encryption, role-based access, GDPR compliance.

3. NON-FUNCTIONAL REQUIREMENTS

3.1. PERFORMANCE

- Load Time: The application will be responsive and load quickly, even on slow network connections
- The application will be able to handle high volumes of data (xxx MB) and support large numbers of users (1000s).
- Zero memory leaks
- Zero constraint issues (warnings and conflict errors)

3.2. SECURITY

- The application will use secure login protocols, such as password protection and two-factor authentication, to ensure the privacy and security of user information. Should have role based authentication for accessing functional modules,
- The application will use secure data storage and encryption to protect sensitive information from unauthorised access.
- The application will comply with relevant data privacy regulations

3.3. USABILITY

- The application will have a user-friendly interface that is easy to navigate and understand.
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3.4. SCALABILITY

- The application should be scalable to accommodate growth and changes in user needs.
- The application should be able to support new features and functionalities as needed.

3.5. RELIABILITY

- The application should be reliable and available at all times, with minimal downtime.
- The application should be able to recover from failures and maintain data consistency in case of errors.
- Zero leaks, zero constraint warnings, zero warnings

3.6. ACCESSIBILITY

- How easily people with the widest range of capabilities can use the system.

4. DETAILED FUNCTIONAL REQUIREMENTS (SAMPLE)

4.1. Borrower:

- 4.1.1. Register and log in using mobile number / email with OTP verification
- 4.1.2. Complete KYC by uploading identity proof, address proof, and income documents
- 4.1.3. Apply for a new loan by selecting loan type, amount, and tenure
- 4.1.4. Track application status in real time — Submitted, Under Review, Approved, Disbursed, Rejected
- 4.1.5. View active loan details including disbursement date, outstanding balance, and interest breakdown
- 4.1.6. Access full EMI schedule with paid, upcoming, and overdue instalments
- 4.1.7. Receive push notifications for due dates, approval decisions, and disbursements
- 4.1.8. Message loan officer in-app for clarifications or document resubmission.

4.2. Loan Officer:

- 4.2.1. View and manage assigned loan applications from a unified dashboard
- 4.2.2. Add comments, request additional documents, or escalate to manage
- 4.2.3. Mark applications as Approved or Recommended for managerial sign-off
- 4.2.4. Access borrower credit history and risk score summary
- 4.2.5. Generate and share loan sanction letters digitally via the app

4.3. Manager:

- 4.3.1. Oversee the full loan portfolio assigned to the branch or region
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- 4.3.2. Approve, reject, or send back loans escalated by officers with remarks
- 4.3.3. View portfolio-level dashboards — total disbursements, collection efficiency, NPA ratio
- 4.3.4. Generate periodic reports (daily, weekly, monthly) exportable as PDF or CSV
- 4.3.5. Configure loan product parameters — interest rates, processing fees, and eligibility criteria

4.4. Admin:

- 4.4.1. Manage user accounts, roles, and permissions across the platform
- 4.4.2. Configure system-wide settings — loan types, document checklists, notification templates
- 4.4.3. Access complete audit trail for all loan actions across all user types

5. SUBMISSIONS

5.1. Code base

5.2. App Video demo

5.3. Memory Profile Screenshot

5.4. Flow diagram